

# Memory and Storage

## Lesson Objectives

By the end of this lesson, students should be able to:

- Differentiate between **primary and secondary storage**.
- Explain the roles of **RAM, ROM, cache, and virtual memory**.
- Identify types of **secondary storage** devices (HDD, SSD, optical, cloud).
- Design a **visual diagram of memory hierarchy** and compare their performance.

## 1 Introduction to Memory and Storage

In computing, **memory and storage** are essential for saving and retrieving data. However, these two terms serve different purposes:

| Term           | Description                                             | Example         |
|----------------|---------------------------------------------------------|-----------------|
| <b>Memory</b>  | Temporary workspace where data is processed (volatile). | RAM             |
| <b>Storage</b> | Long-term data retention (non-volatile).                | Hard Drive, SSD |

### Analogy:

Think of **memory** as your desk (short-term workspace), and **storage** as your filing cabinet (long-term storage).

## 2 Primary vs Secondary Storage

| Feature              | Primary Storage                       | Secondary Storage       |
|----------------------|---------------------------------------|-------------------------|
| <b>Purpose</b>       | Holds data temporarily for processing | Holds data permanently  |
| <b>Speed</b>         | Very fast                             | Slower                  |
| <b>Volatility</b>    | Volatile (data lost when power off)   | Non-volatile            |
| <b>Examples</b>      | RAM, cache, registers                 | HDD, SSD, optical disks |
| <b>Access by CPU</b> | Directly accessible                   | Indirectly via I/O      |

## 3 Types of Primary Memory

### ◆ a) RAM (Random Access Memory)

- Stores data and instructions **currently in use**.
- **Volatile**: Contents are lost when power is off.
- Two main types:
  - **SRAM (Static RAM)**: Faster, used for cache.
  - **DRAM (Dynamic RAM)**: Slower, used for main memory.

💡 Example: When you open MS Word, it loads into **RAM** for quick access.

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### ◆ b) ROM (Read-Only Memory)

- Stores **permanent instructions** used during system boot (e.g., BIOS).
- **Non-volatile**: Data remains even when power is off.
- Types include **PROM**, **EPROM**, and **EEPROM**.

💡 Example: The program that starts your computer when you power it on is stored in ROM.

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### ◆ c) Cache Memory

- **High-speed** memory between the CPU and main memory (RAM).
- Stores frequently used instructions or data to reduce access time.
- Extremely small in size but **very fast**.

🕒 **Cache reduces CPU waiting time** when accessing data repeatedly.

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### ◆ d) Virtual Memory

- Part of the **hard drive used as temporary RAM** when physical memory is full.
- Managed by the **Operating System**.
- Slower than real RAM but prevents crashes when RAM is overloaded.

💡 Example: When running many programs, your computer uses virtual memory (swap file).

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## Types of Secondary Storage

### a) Hard Disk Drive (HDD)

- Uses **magnetic disks** to store data.
- Larger capacity, **slower** access time compared to SSDs.
- Contains moving parts → subject to wear and tear.

 **Typical Capacity:** 500 GB – 2 TB

 **Access Time:** ~5–10 ms

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### b) Solid-State Drive (SSD)

- Stores data in **flash memory** (no moving parts).
- **Faster, quieter, and more durable** than HDDs.
- More expensive per GB.

 **Typical Capacity:** 256 GB – 2 TB

 **Access Time:** ~0.1 ms

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### c) Optical Storage

- Uses **laser technology** to read/write data.
- Includes **CDs, DVDs, Blu-ray discs**.
- Mostly used for **media distribution** or backups.

 Example: Movie DVDs, music CDs.

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### d) Cloud Storage

- Stores data on **remote servers** accessed via the internet.
- Examples: **Google Drive, Dropbox, OneDrive, iCloud**.
- Provides scalability, backup, and global accessibility.

- 🌐 **Advantage:** Access files anywhere.
  - 🔒 **Challenge:** Requires internet & raises privacy concerns.
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## 5 Memory Hierarchy

Memory in a computer is organized in a **hierarchy** based on speed, cost, and capacity.

### ⚙️ Hierarchy Order (Fastest → Slowest):

Registers → Cache → RAM → Secondary Storage → Cloud Storage

| Level | Type                 | Speed     | Capacity            | Cost per Bit |
|-------|----------------------|-----------|---------------------|--------------|
| 1     | <b>Registers</b>     | Fastest   | Smallest            | Highest      |
| 2     | <b>Cache</b>         | Very Fast | Small               | High         |
| 3     | <b>RAM</b>           | Fast      | Moderate            | Medium       |
| 4     | <b>HDD/SSD</b>       | Slower    | Large               | Lower        |
| 5     | <b>Cloud Storage</b> | Slowest   | Virtually Unlimited | Lowest       |

💡 **Principle:**  
As speed increases, capacity decreases — and cost increases.

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## 6 Diagram: Memory Hierarchy

|           |                     |
|-----------|---------------------|
| Registers | (Fastest, Smallest) |
| Cache     |                     |
| RAM       |                     |
| HDD / SSD |                     |

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## 🧠 In-Class Activity

**Activity: Identify and Compare Memory Types**

1. Create a table comparing **RAM, ROM, Cache, Virtual Memory, HDD, SSD, and Cloud Storage**.
  2. Discuss:
    - Which type is the fastest?
    - Which type is permanent?
    - Which is most cost-effective for backups?
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## Mini Project: Memory Hierarchy Visualization

### Objective:

To design a clear **visual representation** of computer memory hierarchy.

### Instructions:

1. Research real-world examples of each memory level (e.g., L1 cache, 16 GB RAM, 1 TB SSD).
2. Create a **visual diagram** (using draw.io, Canva, or PowerPoint).
3. Include:
  - Speed comparisons
  - Capacity ranges
  - Role in the computer system

### Documentation Requirements:

Your report should include:

- A brief explanation of each memory type.
- The hierarchy diagram.
- Reflection: How does memory speed affect computer performance?



### Submission Format:

- File name: memory\_hierarchy\_project.pdf
  - Submit via GitHub repository (week4\_memory\_storage folder).
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## **Git Task**

1. Create a new branch:
  2. `git checkout -b week4-memory-storage`
  3. Add your project files and notes:
  4. `git add memory_hierarchy_project.pdf`
  5. `git commit -m "Added Week 4 memory and storage project"`
  6. Push to your GitHub repo:
  7. `git push origin week4-memory-storage`
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